

Moiz Nafey

+92-302-6402646 • nafeymoiz@gmail.com • linkedin.com/in/moiz-nafey • https://moiznafey.com/

Full-Stack Software Engineer with 3 years building production systems across AI agents, payments, and SaaS platforms. Specialized in Python/FastAPI backends, LangGraph multi-agent orchestration, and AWS cloud infrastructure.

WORK EXPERIENCE

Voya - AI-Powered Travel Planning Platform • London **11/2025 - Present**

Software Engineer - Python/FastAPI, MongoDB, LangGraph, React/TypeScript, AWS, Terraform • Full-time

- Architected dual-path flight and hotel booking pipeline on FlightLogic (LCC WebFare vs. GDS flows) with fare-rules revalidation and PNR reconciliation, processing 2,000+ bookings/day.
- Built three-tier preferences scoring system over a 29K airport resolver, improving flight recommendation relevance and reducing user search-to-book time by 35%.
- Designed agentic AI layer in LangGraph having a Supervisor multi-agent graph (flight, hotel, itinerary agents) featuring capability checks, hallucination recovery, and intent routing; reduced AI response errors by 42% via Langfuse trace.
- Owned Terraform Infrastructure-as-Code across staging/test/prod environments (VPC, ALB, ECS Fargate, ECR, IAM, Secrets Manager, RDS, S3) with GitHub OIDC keyless deployments, cutting deploy time from 14 to 6 minutes.
- Passed SOC 2 readiness security audit by implementing GDPR-compliant field-level PII encryption (Fernet AES-128-CBC + HMAC-SHA256 keys) with masked read views, HMAC-digested step-up OTP, short-lived JWTs, and RBAC.
- Hardened unauthenticated surfaces with per-IP/session sliding-window rate limiter, signed Places image session JWTs, and Sentry alerting, blocking 5K+ malicious requests/month.

The Leading Practice - AI-Powered Marketing Automation Platform • Sheridan, WY **12/2024 - 11/2025**

Software Engineer - (React, ShadCN, Tailwind, Node.js, Express.js, Python, Supabase, CrewAI, Pinecone) • Contractor

- Architected microservices-based multi-tenant SaaS platform with three-level RBAC authorization using Supabase Auth and row-level security, serving 40+ tenants and 5K+ end users.
- Built an AI-powered conversation hub with unified messaging across email, SMS, and live chat, implementing intelligent topic detection, automated categorization, and human-fallback logic for seamless omnichannel communication.
- Engineered Gmail AI automation with RAG-based memory system on Pinecone and Gmail OAuth; implemented two-way email sync via Pub/Sub webhooks, processing 23K+ emails/day with 90-day conversation context retrieval via chunking and embedding strategies.
- Leveraged Model Context Protocol (MCP) framework with CrewAI orchestration to enable dynamic tool execution by AI agents; self-hosted DataForSEO MCP server and implemented meta-agent tool routing for automated SEO analytics report generation, reducing manual report effort by 70%.

SimplifyVms - Talent Management & VMS Platform **05/2023 - 04/2025**

Associate Software Engineer - (React, Django and Flask)

- Worked directly with Fortune 500 clients and engineering managers to ship features aligned with business timelines.
- Played a key role in redesigning and implementing core platform architecture for improved scalability.
- Led the successful onboarding of client (RXO) by independently architecting and implementing a complex real-time integration tailored to their requirements with batch APIs for atomicity and rollbacks.
- Built a rules-based vendor onboarding engine with approval workflows and role-based access.
- Engineered multi-threaded bulk upload system for Marriott, reducing processing time by 60%.
- Collaborated with clients and product teams to deliver features and ensure smooth deployments.

- Wrote unit and integration tests to maintain 80% test coverage across services

SKILLS

Languages: JavaScript, Python, TypeScript

Frontend Frameworks & Libraries: Material UI, React.js, Redux, Redux Toolkit, RTK Query

Backend & Middleware Technologies: Django, Express.js, Fast API, Flask, Node.js

Cloud & DevOps: AWS, CI/CD, CloudWatch, Digital Ocean, Docker, Git, Langfuse, Render, Sentry

Databases: MongoDB, MySQL, Pinecone DB, PostgreSQL, Supabase

AI: CrewAI, LangGraph, LLMs, Multiagent System, Retrieval-Augmented Generation (RAG)

EDUCATION

Bachelors in Computer Science

Lahore University of Management Sciences

08/2019 - 05/2023

Ordinary Levels and Advanced Levels

Lahore Grammar School (JT)

08/2014 - 05/2019